

Step 1: Calculate the Number of Subnets and Hosts

- **Network:** 192.168.1.0/24
- **New Subnet Mask:** /27 (255.255.255.224)

Number of subnets:

- Borrowed bits = $27 - 24 = 3$
- Number of subnets = $2^3 = 8$

Hosts per subnet:

- Host bits = $32 - 27 = 5$
- Total IP addresses per subnet = $2^5 = 32$
- Usable hosts = $32 - 2 = 30$ (excluding the network and broadcast addresses)

Answer:

- **Valid Subnets: 8**
- **Usable Hosts per Subnet: 30**

Subnet	Network Address	Usable Host IP Range	Broadcast Address
1	192.168.1.0/27	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95